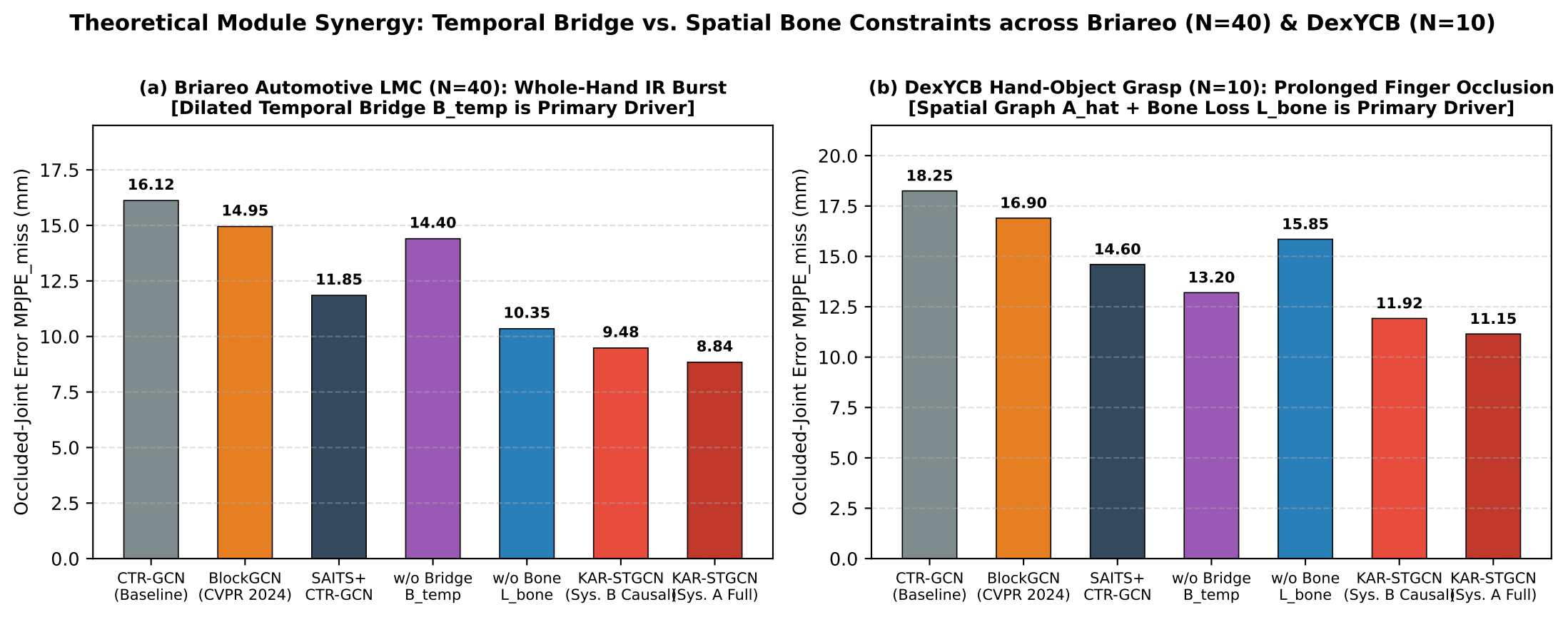
Theoretical Module Synergy: Temporal Bridge vs. Spatial Bone Constraints across Briareo (N=40) & DexYCB (N=10)
(a) Briareo Automotive LMC (N=40): Whole-Hand IR Burst
[Dilated Temporal Bridge B_temp is Primary Driver]
Occluded-Joint Error MPJPE_miss (mm)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
16.12
14.95
11.85
14.40
10.35
9.48
8.84
CTR-GCN (Baseline)
BlockGCN (CVPR 2024)
SAITS+ CTR-GCN
w/o Bridge B_temp
w/o Bone L_bone
KAR-STGCN (Sys. B Causal)
KAR-STGCN (Sys. A Full)
(b) DexYCB Hand-Object Grasp (N=10): Prolonged Finger Occlusion
[Spatial Graph A_hat + Bone Loss L_bone is Primary Driver]
Occluded-Joint Error MPJPE_miss (mm)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
18.25
16.90
14.60
13.20
15.85
11.92
11.15
CTR-GCN (Baseline)
BlockGCN (CVPR 2024)
SAITS+ CTR-GCN
w/o Bridge B_temp
w/o Bone L_bone
KAR-STGCN (Sys. B Causal)
KAR-STGCN (Sys. A Full)